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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2016 Trial Examination

FORM VI

MATHEMATICS EXTENSION 1

Friday 12th August 2016

General Instructions

- Writing time — 2 hours
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 70 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 109 boys

Examiner

SO

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

What are the asymptotes of $y = \frac{2x}{(x+3)(x-1)}$?

- (A) $y = 0$, $x = 1$, $x = -3$
- (B) $y = 0$, $x = -1$, $x = 3$
- (C) $y = 2$, $x = 1$, $x = -3$
- (D) $y = 2$, $x = -1$, $x = 3$

QUESTION TWO

Determine $\lim_{x \rightarrow 0} \left(\frac{\sin x}{3 \tan x} \right)$.

- (A) 0
- (B) $\frac{1}{3}$
- (C) 1
- (D) 3

QUESTION THREE

What is the domain of $f(x) = e^{-\frac{1}{x}}$?

- (A) $x > 0$
- (B) $x \geq 0$
- (C) $x \neq 0$
- (D) all real x

QUESTION FOUR

What is the value of $\sin(\tan^{-1} a)$?

- (A) $\frac{a}{\sqrt{1-a^2}}$
- (B) $\frac{1}{\sqrt{1-a^2}}$
- (C) $\frac{1}{\sqrt{1+a^2}}$
- (D) $\frac{a}{\sqrt{1+a^2}}$

Examination continues next page ...

QUESTION FIVE

The monic quadratic equation with roots $m + n$ and $m - n$ is:

(A) $x^2 - 2mx + m^2 - n^2 = 0$

(B) $x^2 + 2mx + m^2 - n^2 = 0$

(C) $x^2 - 2mx + n^2 - m^2 = 0$

(D) $x^2 + 2mx + n^2 - m^2 = 0$

QUESTION SIX

A function is defined by the following rule:

$$f(x) = \begin{cases} \sin^{-1} x, & \text{for } -1 \leq x < 0 \\ \cos^{-1} x, & \text{for } 0 \leq x \leq 1 \end{cases}$$

What is the value of $f\left(-\frac{1}{2}\right) + f(0)$?

(A) $-\frac{\pi}{6}$

(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{3}$

(D) $\frac{2\pi}{3}$

QUESTION SEVEN

The range R of any particle projected from a point on a level plane at an angle of α to the horizontal with initial speed v is given by $R = \frac{v^2 \sin 2\alpha}{g}$.

A particle is projected at 50° to the horizontal. What other angle of projection would give the same range for this particle?

(A) 25°

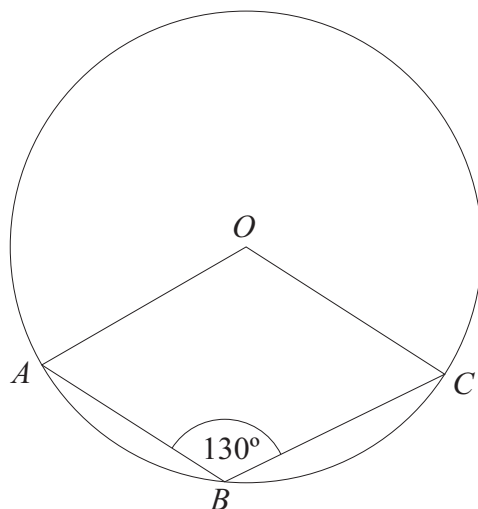
(B) 40°

(C) 80°

(D) 100°

Examination continues overleaf ...

QUESTION EIGHT



The points A , B and C lie on a circle with centre O . If $\angle ABC = 130^\circ$, what is the size of $\angle AOC$?

- (A) 50°
- (B) 65°
- (C) 100°
- (D) 260°

QUESTION NINE

A particle is moving in simple harmonic motion with period 4 and amplitude 3. Which of the following is a possible equation for the velocity of the particle?

- (A) $v = \frac{3\pi}{2} \cos \frac{\pi t}{2}$
- (B) $v = 3 \cos \frac{\pi t}{2}$
- (C) $v = \frac{3\pi}{4} \cos \frac{\pi t}{4}$
- (D) $v = 3 \cos \frac{\pi t}{4}$

QUESTION TEN

Which of the following is a necessary condition if $a^2 > b^2$?

- (A) $a > b$
- (B) $a < b < 0$
- (C) $a > 0 > b$
- (D) $|a| > |b|$

_____ End of Section I _____

Examination continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet. **Marks**

(a) Find the value of $\sin^{-1} \left(\sin \frac{3\pi}{5} \right)$. **1**

(b) Let $A = (4, -3)$ and $B = (8, 5)$. The interval AB is divided internally in the ratio $3 : 1$ by the point $P(x, y)$. Find the values of x and y . **2**

(c) Solve $\frac{5}{3x-2} > 2$. **3**

(d) The acute angle between the two lines $y = \frac{1}{2}x + 1$ and $y = mx + 3$ is $\frac{\pi}{4}$. Find all possible values of the constant m . **3**

(e) Find the general solution of $\cos 2x - \cos x = 2$. **3**

(f) Find the term independent of x in the expansion of $\left(x^2 - \frac{2}{x} \right)^{12}$. **3**

QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

(a) The polynomial $P(x) = 2x^3 + x^2 + ax + 6$ has a zero at $x = 2$.

(i) Determine the value of a .

1

(ii) Find the linear factors of $P(x)$.

2

(iii) Hence, or otherwise, solve $P(x) \geq 0$.

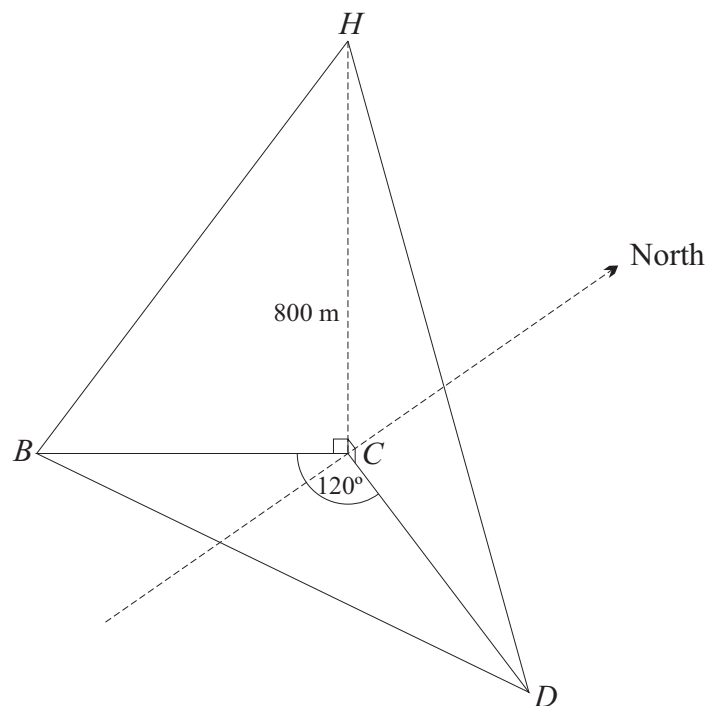
1

(b) Integrate $\int_0^{\frac{1}{\sqrt{3}}} \frac{\sin(\tan^{-1} x)}{1+x^2} dx$ using the substitution $u = \tan^{-1} x$.

3

(c)

2



The diagram above shows a hot air balloon at point H with altitude 800 m. The passengers in the balloon can see a barn and a dam below, at points B and D respectively. Point C is directly below the hot air balloon. From the hot air balloon's position, the barn has a bearing of 250° and the dam has a bearing of 130° , and $\angle BCD = 120^\circ$. The angles of depression to the barn and the dam are 50° and 30° respectively.

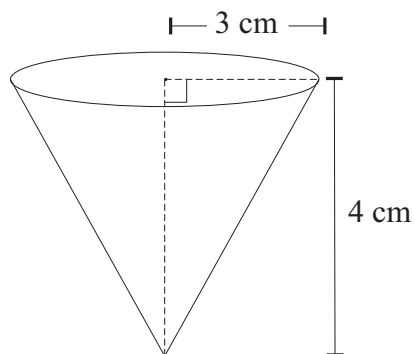
How far is the barn from the dam, to the nearest metre?

(d) Prove by induction that $(x + y)$ is a factor of $x^{2n} - y^{2n}$, for all integers $n \geq 1$.

3

(e)

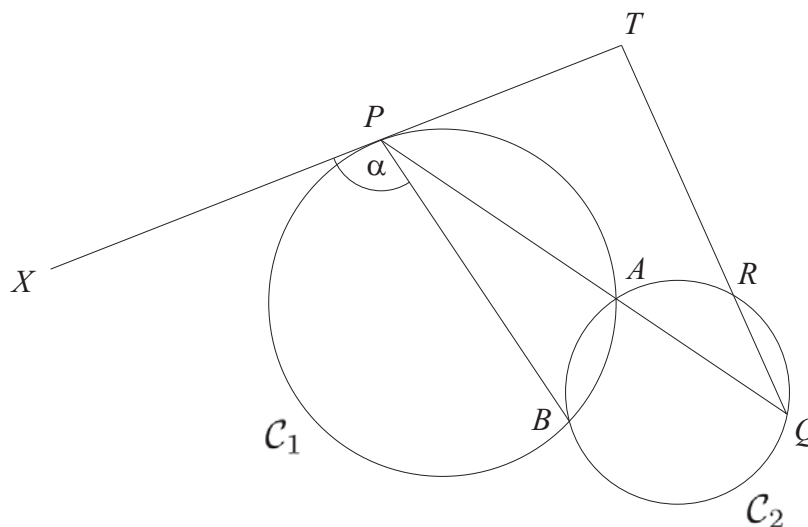
3



The diagram above shows a vessel in the shape of a cone of radius 3 cm and height 4 cm. Water is poured into it at the rate of $10 \text{ cm}^3/\text{s}$. Find the rate at which the water level is rising when the depth is 2 cm.

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.**Marks**

(a)



Two circles C_1 and C_2 intersect at A and B . A line through A meets the circles at P and Q respectively. A tangent is drawn from an external point T to touch the circle C_1 at P . The line TQ intersects C_2 at R .

(i) Given $\angle XPB = \alpha$, show that $\angle BRQ = 180^\circ - \alpha$, giving reasons. 2

(ii) Hence show that $PTRB$ is a cyclic quadrilateral. 1

(b) Consider the parabola $x^2 = 4ay$ with focus S . The normal at $P(2ap, ap^2)$ meets the y -axis at R and $\triangle SPR$ is equilateral.

(i) Show that the equation of the normal at P is $x + py = 2ap + ap^3$. 1

(ii) Write down the coordinates of R . 1

(iii) Prove that SP is equal in length to the latus rectum, that is $4a$ units. 3

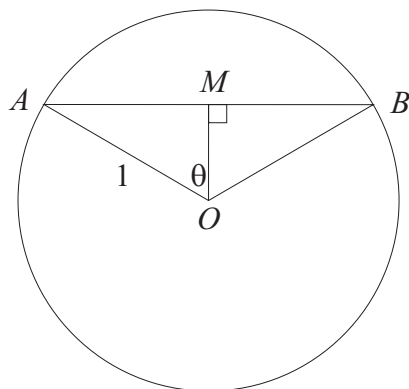
(c) (i) Show that $\frac{d}{dx}(x \ln x) = 1 + \ln x$. 1

(ii) A particle is moving in a straight line. At time t seconds its position is x cm and its velocity is v cm/s. Initially $x = 1$ and $v = 2$. The acceleration a of the particle is given by the equation

$$a = 1 + \ln x.$$

Find the velocity v in terms of x . Be careful to explain why v is always positive. 2

(d)



The circle above has radius 1 unit and the major arc joining A and B is twice as long as the chord AB . The point M lies on AB such that $AB \perp OM$. Let $\angle AOM = \theta$ where $0 < \theta < \frac{\pi}{2}$.

- (i) Show that the length of the major arc satisfies the equation

1

$$\pi - \theta = 2 \sin \theta.$$

- (ii) Let $\theta_0 \doteq 1.5$ be a first approximation of θ . Use two applications of Newton's method to find a better approximation of θ .

2

- (iii) Use your answer to part (ii) to find the approximate length of the chord AB .

1

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) The mass M of a radioactive isotope is given by the equation $M = M_0 e^{-kt}$, where M_0 is the initial mass and k is a constant. The mass satisfies the equation $\frac{dM}{dt} = -kM$.

(i) If the half-life of this radioactive isotope is T , show that $k = \frac{\log_e 2}{T}$. 1

- (ii) A naturally occurring rock contains two radioactive isotopes X and Y . The half-lives of isotope X and isotope Y are T_X and T_Y respectively, where $T_X > T_Y$. Initially the mass of isotope Y is four times that of isotope X . 3

Show that the rock will contain the same mass of both isotopes at time

$$\frac{2T_X T_Y}{T_X - T_Y}.$$

- (b) Sketch the graph of $y = \frac{|x|}{x}$. 1

- (c) Consider the function $f(x) = \sin^{-1} \left(\frac{x^2 - 1}{x^2 + 1} \right)$.

- (i) Find the domain of $f(x)$. 1

- (ii) Show that $f'(x) = \frac{2x}{|x|(x^2 + 1)}$. 2

- (iii) Determine the values of x for which $f(x)$ is increasing. 1

- (iv) Using part (b), explain the behaviour of $f'(x)$ as $x \rightarrow 0^+$ and $x \rightarrow 0^-$. 1

- (v) Draw a neat sketch of $y = f(x)$, indicating any intercepts with the axes and any asymptotes. 2

- (vi) Give the largest possible domain containing $x = 1$ for which $f(x)$ has an inverse function. Let this inverse function be $f^{-1}(x)$. 1

- (vii) Sketch $y = f^{-1}(x)$ on your original graph. 1

- (viii) Find the equation of $f^{-1}(x)$. 1

————— End of Section II —————

END OF EXAMINATION

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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

EXTENSION MATHEMATICS SOLUTIONS

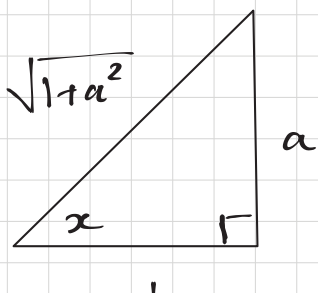
MULTIPLE CHOICE

1. Vertical asymptotes : $(x+3)(x-1)=0$
 $x=-3$ and $x=1$ A

Horizontal asymptote: $y=0$

2. $\lim_{x \rightarrow 0} \frac{\sin x}{x} \times \frac{x}{\tan x} \times \frac{1}{3} = \frac{1}{3}$ B

3. $f(x) = e^{-\frac{1}{x}} \quad x \neq 0$ C

4.  Let $x = \tan^{-1} a$
 $\tan x = a$
 $\sin x = \frac{a}{\sqrt{1+a^2}}$ D

5. $(x-(m+n))(x-(m-n)) = 0$
 $x^2 - 2mx + m^2 - n^2 = 0$ A

6. $f(-\frac{1}{2}) + f(0)$
 $= \sin^{-1}(-\frac{1}{2}) + \cos^{-1}(0)$
 $= -\frac{\pi}{6} + \frac{\pi}{2}$
 $= \frac{\pi}{3}$ C

7. $\sin(2 \times 50^\circ) = \sin 100^\circ = \sin 80^\circ = \sin(2 \times 40^\circ)$ B

8. Reflex $\angle AOC = 260^\circ$ (angle at the centre is twice the angle at the circumference)
 $\therefore \angle AOC = 100^\circ$ C

9. $T = \frac{2\pi}{n}$
 $= \frac{\pi}{2}$

$x = 3 \sin \frac{\pi t}{2}$
 $v = \frac{3\pi}{2} \cos \frac{\pi t}{2}$

A

10. A sufficient
 B neither necessary nor sufficient
 C sufficient
 D necessary and sufficient

1	2	3	4	5	6	7	8	9	10
A	B	C	D	A	C	B	C	A	D

Q11

$$(a) \sin^{-1}\left(\sin\left(\frac{3\pi}{5}\right)\right)$$

$$= \sin^{-1}\left(\sin\frac{2\pi}{5}\right)$$

$$= \frac{2\pi}{5} \quad \checkmark$$

$$(b) \quad \begin{array}{cc} A & B \\ (4, -3) & (8, 5) \\ 3 & 1 \end{array}$$

$$x = \frac{3 \times 8 + 1 \times 4}{4}$$

$$= \frac{28}{4}$$

$$= 7 \quad \checkmark$$

$$y = \frac{3 \times 5 - 1 \times 3}{4}$$

$$= \frac{12}{4}$$

$$= 3 \quad \checkmark$$

$$S_0 (x, y) = (7, 3)$$

$$(c) \quad \frac{5}{3x-2} > 2$$

$$5(3x-2) > 2(3x-2)^2 \quad \checkmark$$

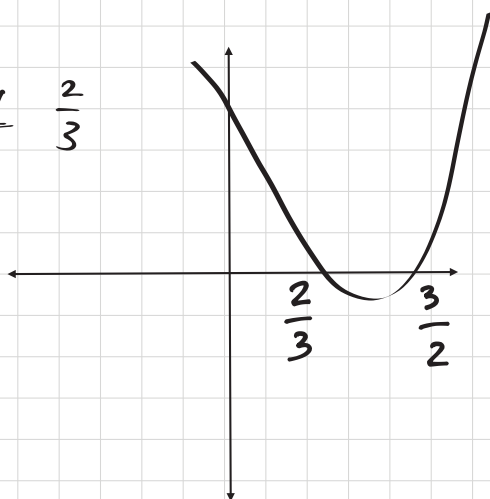
$$2(3x-2)^2 - 5(3x-2) < 0$$

$$(3x-2)(6x-9) < 0 \quad \checkmark$$

$$3(3x-2)(2x-3) < 0$$

$$S_0 \quad \frac{2}{3} < x < \frac{3}{2} \quad \checkmark$$

$$x \neq \frac{2}{3}$$



$$(d) \quad \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$1 = \left| \frac{m_1 - \frac{1}{2}}{1 + \frac{m_1}{2}} \right| \quad \checkmark$$

$$1 + \frac{m_1}{2} = m_1 - \frac{1}{2}$$

$$-\frac{m_1}{2} = -\frac{3}{2}$$

$$m_1 = 3 \quad \checkmark$$

$$\text{or} \quad 1 + \frac{m_1}{2} = -m_1 + \frac{1}{2}$$

$$\frac{3m_1}{2} = -\frac{1}{2}$$

$$m_1 = -\frac{1}{3} \quad \checkmark$$

So the possible values of the constant m are 3 and $-\frac{1}{3}$.

$$(e) \quad \cos 2x - \cos x = 2$$

$$2\cos^2 x - 1 - \cos x - 2 = 0 \quad \checkmark$$

$$2\cos^2 x - \cos x - 3 = 0$$

$$(2\cos x - 3)(\cos x + 1) = 0$$

$$\cos x = \frac{3}{2}$$

↓

no solution

$$\cos x = -1 \quad \checkmark$$

$$x = -\pi + 2n\pi, \text{ where } n \text{ is an integer} \quad \checkmark$$

$$(f) \quad \text{Let the term be } {}^{12}C_k (x^2)^{12-k} \left(-\frac{2}{x}\right)^k$$

$$= {}^{12}C_k \times x^{24-2k} \times (-1)^k \times 2^k \times x^{-k}$$

$$= {}^{12}C_k \times x^{24-3k} \times (-1)^k \times 2^k \quad \checkmark$$

$$\text{For the term independent of } x, \quad 24 - 3k = 0$$

$$k = 8 \quad \checkmark$$

$$\text{So } {}^{12}C_8 \times x^0 \times (-1)^8 \times 2^8$$

$$= {}^{12}C_8 \times 2^8 \quad \checkmark$$

$$= 495 \times 256$$

$$= 126720$$

Q.N 12

(a)(i) $P(2) = 0$

$$0 = 16 + 4 + 2a + 6$$

$$2a = -26$$

$$a = -13 \quad \checkmark$$

(ii)
$$\begin{array}{r} 2x^2 + 5x - 3 \\ x-2 \overline{) 2x^3 + x^2 - 13x + 6} \end{array} \quad \checkmark$$

$$2x^3 - 4x^2$$

$$5x^2 - 13x$$

$$5x^2 - 10x$$

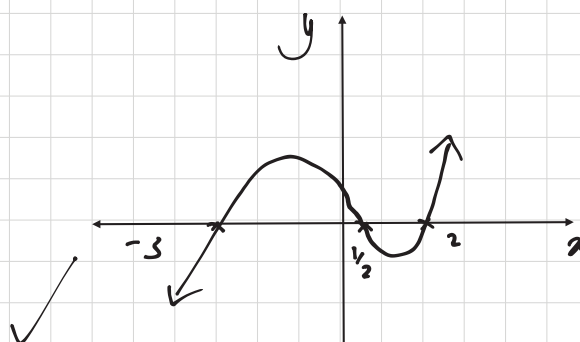
$$-3x + 6$$

$$-3x + 6$$

$$2x^2 + 5x - 3 = (2x-1)(x+3)$$

$$\text{so } P(x) = (x-2)(2x-1)(x+3) \quad \checkmark$$

(iii) $P(x) \geq 0$
 $-3 \leq x \leq \frac{1}{2} \text{ or } x \geq 2$



(b)
$$\int_0^{\frac{1}{\sqrt{3}}} \frac{\sin(\tan^{-1}x)}{1+x^2} dx$$

$$= \int_0^{\frac{\pi}{6}} \sin u \, du \quad \checkmark$$

$$= -[\cos u]_0^{\frac{\pi}{6}} \quad \checkmark$$

$$= -\left(\frac{\sqrt{3}}{2} - 1\right)$$

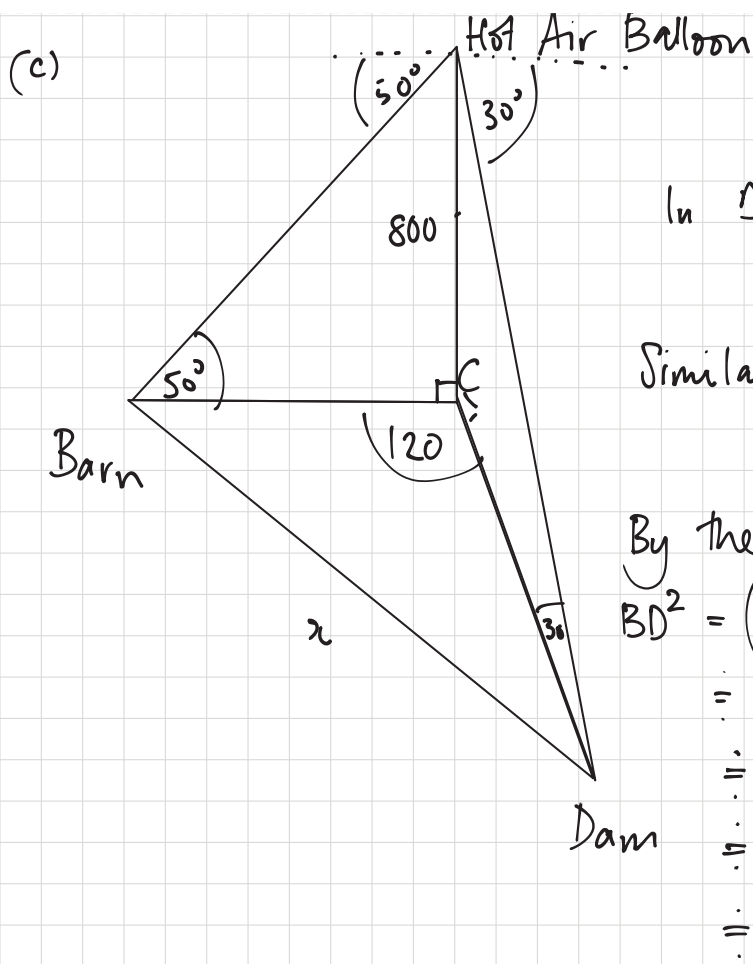
$$= 1 - \frac{\sqrt{3}}{2} \quad \checkmark$$

$$= \frac{2 - \sqrt{3}}{2}$$

$$u = \tan^{-1}x$$

$$du = \frac{dx}{1+x^2}$$

When $x = \frac{1}{\sqrt{3}} \quad u = \frac{\pi}{6}$
 $x = 0 \quad u = 0$



$$\text{In } \triangle BCD, \tan 50^\circ = \frac{800}{BC}$$

$$BC = 800 \cot 50^\circ$$

$$\text{Similarly in } \triangle DCD, \tan 30^\circ = \frac{800}{DC}$$

$$DC = 800 \cot 30^\circ$$

By the cosine rule:

$$BD^2 = (800^2 \cot^2 50^\circ + 800^2 \cot^2 30^\circ - 2 \cdot 800 \cot 50^\circ \cdot 800 \cot 30^\circ \cos 120^\circ)$$

$$= 800^2 \times 5.157$$

$$= 3300488$$

$$= 1816.722...$$

$$= 1817m$$

(d) Prove $x^{2n} - y^{2n}$ has $(x+y)$ as a factor for all integers $n \geq 1$.

① For $n=1$: $x^2 - y^2 = (x+y)(x-y)$, which has $(x+y)$ as a factor

② Assume true for $n=k$:

$$x^{2k} - y^{2k} = m(x+y) \quad (*) \text{ where } m \text{ is an expression in } x \text{ \& } y$$

③ Prove true for $n=k+1$:

$$\text{From } (*) \quad x^{2k} = m(x+y) + y^{2k}$$

$$x^{2k+2} - y^{2k+2}$$

$$= x \cdot x^{2k} - y \cdot y^{2k}$$

$$= x^2 [m(x+y) + y^{2k}] - y^2 \cdot y^{2k}$$

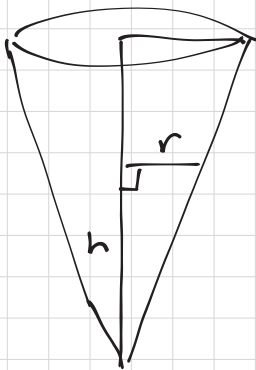
$$= mx^2(x+y) + x^2 y^{2k} - y^2 \cdot y^{2k}$$

$$= mx^2(x+y) + y^{2k}(x^2 - y^2)$$

$$= x+y [mx^2 + y^{2k}(x-y)]$$

So from parts (2) & (3) and by mathematical induction $(x+y)$ is a factor of $x^{2n} - y^{2n}$ for all integers $n \geq 1$.

(e)



$$\frac{r}{h} = \frac{3}{4}$$

$$r = \frac{3h}{4}$$

$$V = \frac{1}{3} \pi \left(\frac{3h}{4} \right)^2 h$$

$$= \frac{3h^3 \pi}{16}$$

$$\frac{dV}{dh} = \frac{9h^2 \pi}{16}$$

$$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$10 = \frac{9 \times 4 \times \pi}{16} \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{40}{9\pi} \text{ cm/s}$$

∴ the water is rising at a rate of $\frac{40}{9\pi}$ cm/s when the depth of the cone is 2cm

Q13

(a)(i) $\angle XPB = \alpha$ Join Points A and B, and B and R.

$\angle PAB = \alpha$ (alternate segment theorem) ✓

$\angle BAO = 180^\circ - \alpha$ (straight line)

$\angle BRQ = 180^\circ - \alpha$ (angles standing on the same arc) ✓

(ii) $\angle TRB = \alpha$ (straight angle)

$\angle XPT = \angle TRB$

∴ PTRB is a cyclic quadrilateral

(Exterior angle is equal to the opposite interior angle) ✓

(b)(i)

$$\frac{dy}{dx} = \frac{2ap}{2a}$$

$$= p$$

so gradient of the normal is $-\frac{1}{p}$

equation of the normal :

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

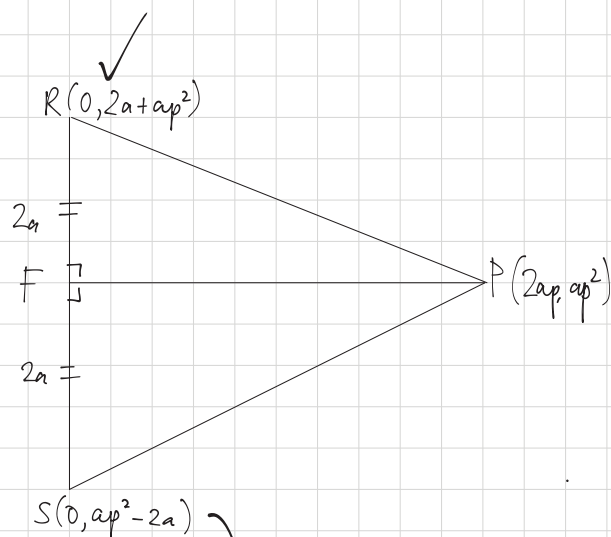
$$x + py = 2ap + ap^3$$

(ii) $R(0, 2a + ap^2)$

(iii) Let F be foot of the perpendicular from P to the y-axis.

Then $FR = FS = 2a$

(Δ SRP is isosceles with altitude FP)

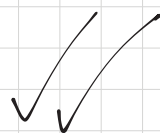


Hence $SR = 4a$ and since Δ SPR is equilateral

$$SP = SR$$

$$= 4a$$

= length of latus rectum.



(c) (i) $\frac{d}{dx}(x \ln x) = \ln x + \frac{1}{x} \times x$

$$= \ln x + 1$$



(ii) $a = 1 + \ln x$

$$\frac{1}{2} v^2 = x \ln x + c$$

at $v=2, x=1$ so $c=2$

$$\frac{1}{2} v^2 = x \ln x + 2$$

$$v^2 = 2x \ln x + 4$$

$$v = \pm \sqrt{2x \ln x + 4}$$

$\ln x$ is always increasing
so at $t=0, a > 0$

so v is always positive

$$\therefore v = \sqrt{2x \ln x + 4}$$

(d) (i) Let $AM = x$

chord $AB = 2x$

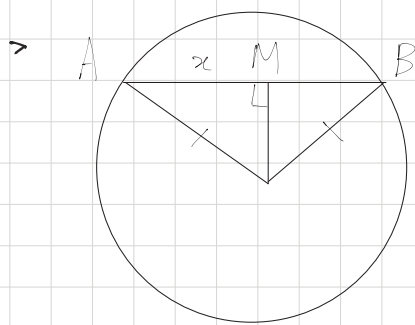
major arc $AB = 4x$

$$x = \sin \theta$$

$$\text{So } AB = 2 \sin \theta$$

$$\text{Major arc } AB = 2\pi - 2\theta$$

$$\text{So } \begin{aligned} 2\pi - 2\theta &= 4 \sin \theta \\ \pi - \theta &= 2 \sin \theta \end{aligned} \quad \checkmark$$



$$(ii) \quad 2 \sin \theta + \theta - \pi = 0$$

$$\text{So } f(\theta) = 2 \sin \theta + \theta - \pi$$

$$f'(\theta) = 2 \cos \theta + 1 \quad \checkmark$$

$$\theta_1 = \theta_0 - \frac{f(\theta)}{f'(\theta)}$$

$$= 1.5 - \frac{(2 \sin \theta + \theta - \pi)}{2 \cos \theta + 1}$$

$$= 1.19$$

$$\theta_2 = 1.24433... \quad \checkmark$$

$$(iii) \quad AB = 2 \sin \theta$$

$$= 2 \times \sin(1.24433...)$$

$$= 1.89436... \quad \checkmark$$

Q14

$$(a) (i) \frac{1}{2} M_0 = M_0 e^{-kT}$$

$$\frac{1}{2} = e^{-kT}$$

$$\ln\left(\frac{1}{2}\right) = -kT$$

$$k = \frac{\ln 2}{T} \quad \checkmark$$

$$(ii) X = X_0 e^{-k_x t}$$

$$Y = Y_0 e^{-k_r t} = 4X_0 e^{-k_r t}$$

$$4X_0 e^{-k_r t} = X_0 e^{-k_x t} \quad \checkmark$$

$$\frac{4}{e^{k_r t}} = \frac{1}{e^{k_x t}}$$

$$4 = \frac{e^{k_r t}}{e^{k_x t}}$$

$$4 = e^{t(k_r - k_x)}$$

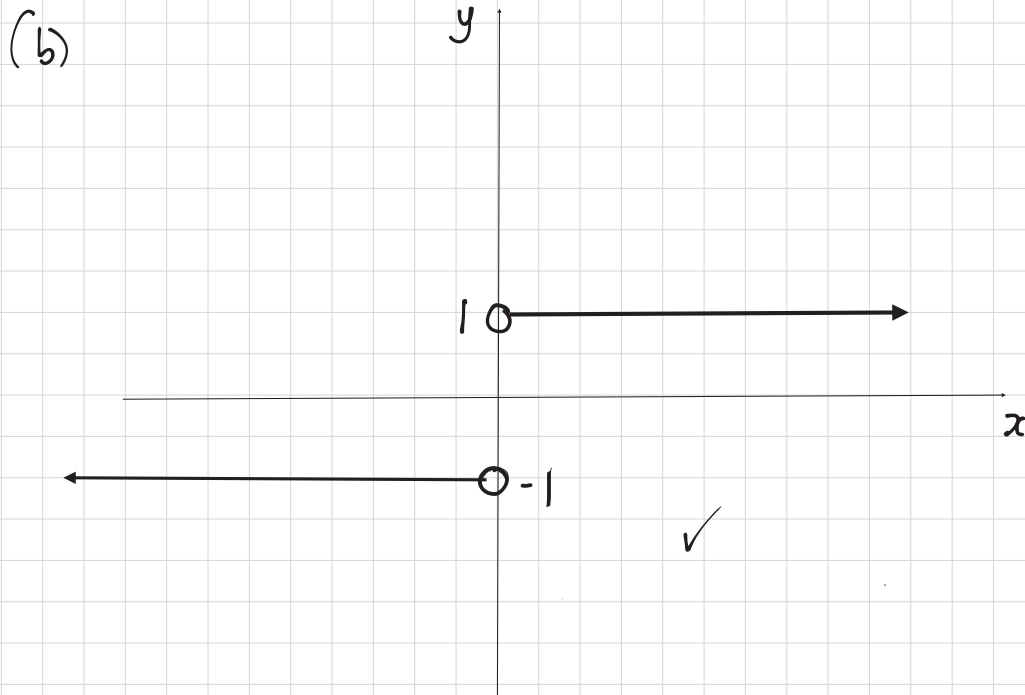
$$2 \ln 2 = t(k_r - k_x)$$

$$2 \ln 2 = t \left(\frac{\ln 2}{T_r} - \frac{\ln 2}{T_x} \right) \quad \checkmark$$

(using part (i))

$$2 = t \left(\frac{T_x - T_r}{T_x T_r} \right)$$

$$t = \frac{2 T_x T_r}{T_x - T_r} \quad \checkmark$$



(c) (i) $f(x) = \sin^{-1}\left(\frac{x^2-1}{x^2+1}\right)$

Domain: $-1 \leq \left(\frac{x^2-1}{x^2+1}\right) \leq 1$

$$-x^2-1 \leq x^2-1 \leq x^2+1$$

$$\begin{aligned} -x^2-1 &\leq x^2-1 & \text{and} & & x^2-1 &\leq x^2+1 \\ -2x^2 &\leq 0 & & & 0 &\leq 2 \\ -x^2 &\leq 0 & & & \text{true for all } x \end{aligned}$$

So domain is all real x

(ii) Let $y = \sin^{-1} u$

$$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{1}{\sqrt{1-\left(\frac{x^2-1}{x^2+1}\right)^2}} \times \frac{4x}{(x^2+1)^2}$$

$$= \frac{(x^2+1) \times 4x}{\sqrt{x^4+2x^2+1-(x^4-2x^2+1)}} \times (x^2+1)^2$$

$$= \frac{4x}{\sqrt{4x^2} \times (x^2+1)}$$

$$\begin{aligned} u &= x^2-1 & v &= x^2+1 \\ u' &= 2x & v' &= 2x \end{aligned}$$

$$\begin{aligned} \frac{du}{dx} &= \frac{2x(x^2+1)-2x(x^2-1)}{(x^2+1)^2} \\ &= \frac{2x(x^2+1-x^2+1)}{(x^2+1)^2} \\ &= \frac{4x}{(x^2+1)^2} \end{aligned}$$

$$= \frac{4x}{2|x| \times (x^2+1)} \quad (\text{by definition } \sqrt{x^2} = |x|)$$

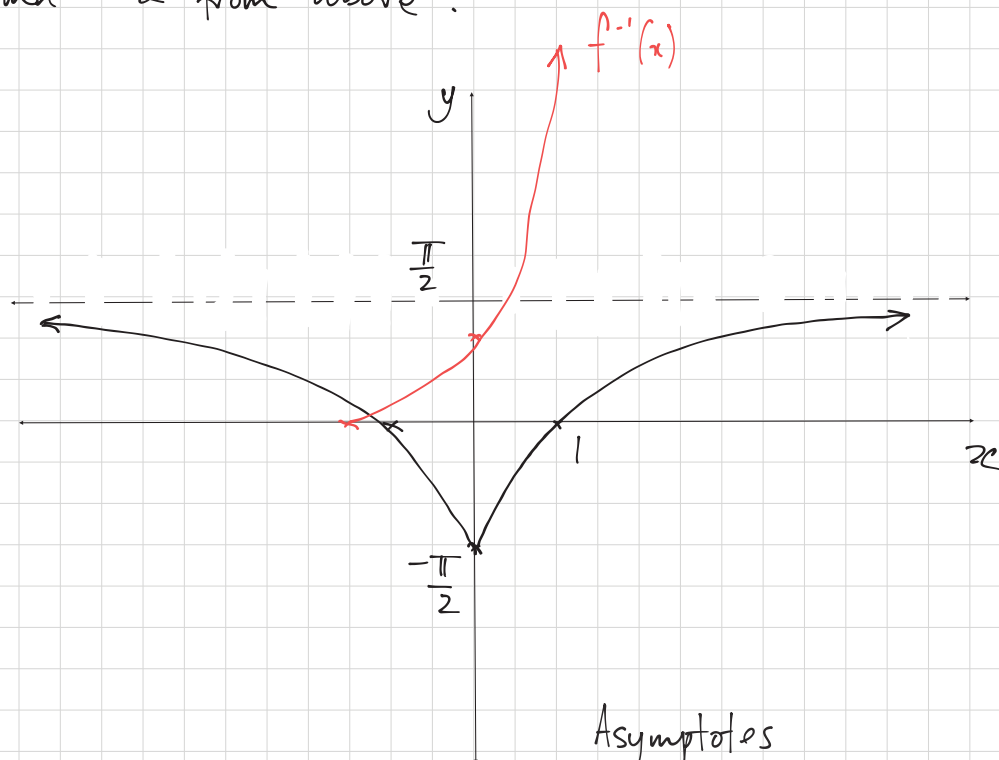
$$= \frac{2x}{|x| \times (x^2+1)} \quad \checkmark$$

(iii) Increasing when $\frac{dy}{dx} > 0$
 $x > 0 \quad \checkmark$

$$(iv) \lim_{x \rightarrow 0^+} f'(x) = 2$$

$$\lim_{x \rightarrow 0^-} f'(x) = -2 \quad \checkmark$$

The gradient approaches +2 from below and -2 from above.



x-intercept at $f(x) = 0$

$$x^2 - 1 = 0$$

$$x = \pm 1$$

Asymptotes

$$\text{as } x \rightarrow \infty \quad f(x) \rightarrow \sin^{-1} 1$$

$$\rightarrow \frac{\pi}{2}$$

$$\text{as } x \rightarrow -\infty \quad f(x) \rightarrow \sin^{-1}(-1)$$

$$\rightarrow -\frac{\pi}{2}$$

Horizontal asymptote: $y = \frac{\pi}{2}$

(vi) $x \geq 0$ ✓

(vii) See graph ✓

(viii) $x = \sin^{-1}\left(\frac{y^2-1}{y^2+1}\right)$

$$\sin x = \frac{y^2-1}{y^2+1}$$

$$\sin x (y^2+1) = y^2-1$$

$$y^2 \sin x + \sin x = y^2-1$$

$$y^2 (1-\sin x) = 1+\sin x$$

$$y^2 = \frac{1+\sin x}{1-\sin x}$$

$$y = \sqrt{\frac{1+\sin x}{1-\sin x}}$$

$$\therefore f^{-1}(x) = \sqrt{\frac{1+\sin x}{1-\sin x}}$$